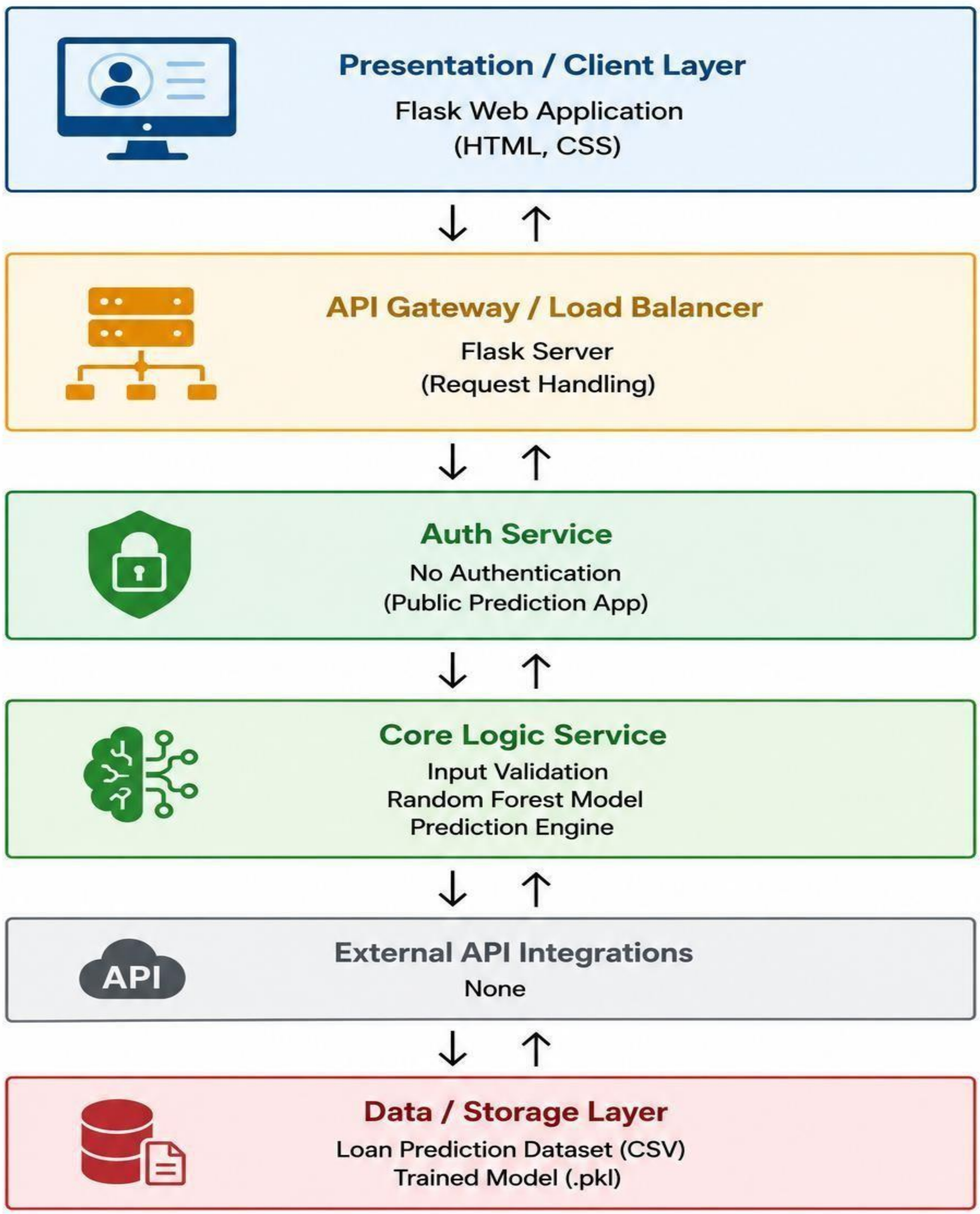
Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Smart Lender – Loan Approval Prediction System follows a simple three-layer architecture consisting of the presentation layer, application layer, and data layer. The presentation layer allows users to interact with the system through a web interface developed using HTML, CSS, and Flask. The application layer processes user requests, validates the entered information, and uses a trained Random Forest machine learning model to predict loan approval. The data layer contains the loan prediction dataset used during model training and the trained Pickle model used for inference. This architecture ensures smooth communication between system components while providing accurate, efficient, and reliable loan eligibility predictions. The modular design also allows future enhancements and simplifies maintenance and deployment.

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides a user-friendly interface where applicants enter loan details and view the loan eligibility prediction result.	HTML, CSS, JavaScript, Flask Templates
API Gateway / Application Layer	Receives user requests, validates input, and forwards the data to the machine learning prediction service.	Flask, Python
Core Logic Service	Performs data preprocessing, feature encoding, and executes the trained XGBoost model to predict loan eligibility.	Python, Scikit-learn, XGBoost, Pandas, NumPy
External API Integrations	Supports future integration with banking systems, credit verification services, or notification services if required.	REST APIs (Optional)